

DAUPHIN, MB, Canada

WMO: 715500

Lat: 51.1003N

Lon: 100.0569W

Elev: 999

StdP: 14.17

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.8	-20.3	-33.4	0.9	-24.5	-28.9	1.1	-19.9	29.3	12.0	26.9	14.5	7.9	250	0.696

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.0	85.9	69.4	82.7	67.0	79.8	65.6	72.2	81.9	69.9	79.3	67.8	76.7	11.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.1	110.9	77.9	66.5	101.4	75.2	64.3	93.5	73.3	36.7	81.7	34.6	79.4	32.8	76.9	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.9	23.0	20.5	-32.3	92.5	4.4	3.4	-35.4	94.9	-38.0	96.9	-40.4	98.8	-43.6	101.3
			-32.5	77.0	4.3	2.7	-35.6	78.9	-38.1	80.5	-40.6	82.0	-43.7	83.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	37.0	4.2	7.6	20.3	37.6	50.6	61.4	66.3	64.3	55.2	40.9	23.9	9.4
	DBStd	24.77	15.27	13.95	14.35	11.47	9.35	7.06	5.75	6.28	8.19	8.98	12.49	14.49
	HDD50	6420	1421	1187	921	393	108	4	0	0	39	306	783	1258
	HDD65	10456	1886	1607	1385	822	456	148	54	89	307	747	1232	1723
	CDD50	1665	0	0	0	21	126	347	506	444	195	25	1	0
	CDD65	225	0	0	0	0	8	40	95	68	14	0	0	0
	CDH74	2494	0	0	0	11	147	429	911	775	211	10	0	0
	CDH80	617	0	0	0	1	31	101	219	217	47	1	0	0
Wind	WSAvg	9.4	9.7	9.6	9.3	9.9	10.4	9.2	8.2	8.3	9.1	9.7	9.7	9.5
Precipitation	PrecAvg	20.2	0.8	0.7	1.1	1.3	2.2	3.1	2.8	2.6	2.5	1.3	0.9	0.9
	PrecMax	28.6	2.0	1.7	3.0	3.5	5.2	7.9	7.8	7.1	6.8	3.9	2.4	1.8
	PrecMin	11.9	0.2	0.2	0.1	0.0	0.2	0.3	1.1	0.2	0.3	0.1	0.0	0.3
	PrecStd	3.8	0.5	0.4	0.7	0.9	1.3	1.9	1.5	1.8	1.6	1.1	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	40.3	40.7	55.4	74.8	83.9	87.9	89.1	90.8	85.4	73.4	58.5	44.5
		MCWB	36.0	36.0	44.4	55.2	61.2	69.9	74.3	70.6	65.9	57.5	47.5	38.0
	2%	DB	36.5	36.3	48.6	67.5	78.1	82.7	85.2	85.5	80.0	65.9	50.6	38.4
		MCWB	33.2	33.2	41.4	51.3	57.8	65.5	70.9	68.8	63.6	52.0	42.5	34.0
	5%	DB	31.9	33.0	42.9	62.0	73.0	78.8	82.2	81.6	75.3	60.4	44.6	33.8
		MCWB	29.4	30.8	37.9	48.7	55.8	64.1	68.8	67.3	60.9	50.1	38.7	30.4
	10%	DB	26.8	28.7	38.6	56.3	68.4	75.4	79.2	78.2	70.4	55.3	40.1	29.0
		MCWB	24.9	26.8	35.2	45.7	53.8	62.2	67.3	65.4	58.7	47.3	36.1	27.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.3	36.4	46.5	56.4	64.4	73.8	76.5	74.5	69.2	58.8	48.3	37.7
		MCDB	39.7	39.5	53.4	71.1	77.5	83.5	85.5	84.9	78.8	70.4	56.9	43.9
	2%	WB	33.5	33.5	41.7	52.2	60.5	68.9	73.1	71.1	65.6	54.2	43.3	34.4
		MCDB	36.4	35.9	47.9	65.4	73.1	78.6	82.3	81.7	76.1	63.0	49.7	38.5
	5%	WB	29.4	30.7	38.2	49.3	57.7	66.1	70.7	69.0	62.8	51.0	39.1	30.6
		MCDB	32.0	33.0	42.7	60.5	69.8	74.8	79.6	78.8	72.8	58.3	43.4	33.6
	10%	WB	24.8	26.8	35.4	46.2	55.0	63.9	68.6	66.8	59.7	48.3	36.1	27.3
		MCDB	26.5	28.7	38.4	56.1	65.8	72.8	77.0	75.9	68.9	55.1	39.9	29.3
Mean Daily Temperature Range	5% DB	MDBR	16.4	17.6	18.1	21.1	22.8	20.2	21.0	22.6	21.7	17.7	14.9	15.5
		MCDBR	18.3	17.6	20.1	30.4	30.3	25.3	24.8	27.9	30.0	27.2	21.4	18.8
		MCWBR	16.0	15.0	14.8	17.5	15.4	12.8	13.5	14.5	16.1	16.4	14.9	15.5
	5% WB	MCDBR	18.0	16.9	19.3	28.0	25.9	21.5	22.9	24.8	27.0	24.5	20.1	18.1
		MCWBR	15.9	14.7	14.4	16.8	14.8	12.3	13.8	14.4	15.9	15.4	14.7	15.1
Clear-Sky Solar Irradiance	taub	0.263	0.267	0.305	0.337	0.350	0.366	0.376	0.372	0.340	0.293	0.263	0.243	
	taud	2.253	2.282	2.297	2.395	2.417	2.400	2.394	2.398	2.473	2.566	2.467	2.343	
	Ebn at Noon	241	277	284	287	287	282	277	272	270	268	248	236	
	Edh at Noon	21	27	33	34	35	36	35	34	28	21	17	16	
All-Sky Solar Radiation	RadAvg	402	775	1226	1609	1778	1840	1953	1602	1163	673	359	285	
	RadStd	31	51	87	112	156	134	91	128	100	88	49	27	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	+3.20	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	-0.90	N/A	N/A	N/A	N/A

Nomenclature: See separate page